Circulation: Cardiovascular Quality and Outcomes

ORIGINAL ARTICLE

Cluster-Randomized Trial to Evaluate a Centralized Clinical Pharmacy Service in Private Family Medicine Offices

See Editorial by Pearson

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BACKGROUND: The use of clinical pharmacists in primary care has improved the control of several chronic cardiovascular conditions. However, many private physician practices lack the resources to implement team-based care with pharmacists. The purpose of this study was to evaluate whether a centralized, remote, clinical pharmacy service could improve guideline adherence and secondary measures of cardiovascular risk in primary care offices in rural and small communities.

METHODS AND RESULTS: This study was a prospective trial in 12 family medicine offices cluster randomized to either the intervention or usual care. The intervention was delivered for 12 months, and subjects had research visits at baseline and 12 months. The primary outcome was adherence to guidelines, and secondary outcomes included changes in key cardiovascular risk factors and preventative health measures. We enrolled 302 subjects. There was no improvement in the Guideline Advantage score from baseline to 12 months in the control group (64.7% versus 63.1%, respectively; P=0.21). There was a statistically significant improvement in the intervention group from 63.3% at baseline to 67.8% at 12 months (P=0.02). The estimated benefit of the intervention was 5.0%±2.4% (95% confidence interval=−0.5% to 10.4%; P=0.07). Several criteria were significantly better for intervention subjects, including appropriate statin therapy (P<0.001), body mass index, screening (P<0.001), and alcohol screening (P<0.001). Only 13.7% of subjects with diabetes mellitus had hemoglobin A1c at goal at baseline, and this increased to 30.8% and 21.0% in the intervention and control group, respectively, at 12 months (P=0.10).

CONCLUSIONS: The centralized, remote pharmacist intervention was successfully implemented. The improvements in outcomes were modest, in part because of higher than expected baseline guideline adherence. Future studies of this model should focus on patients with uncontrolled conditions at high risk for cardiovascular events.

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Key Words: cardiovascular diseases ■ diabetes mellitus ■ guideline adherence ■ risk factors

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WHAT IS KNOWN

- Involving pharmacists in team-based care has been shown to significantly improve care when compared with usual care.
- Remote telemedicine services have been shown to improve cardiovascular risk factors, but most studies have involved integrated health systems.

WHAT THE STUDY ADDS

- The centralized, remote pharmacist intervention was successfully implemented.
- Physicians completely accepted 88.8% of pharmacist recommendations and 6% of recommendations with minor modifications (95% overall).
- Despite higher than expected baseline guideline adherence, the intervention significantly improved overall guideline adherence and several individual criteria.

More than 2200 Americans die of cardiovascular disease every day or 1 death every 39 seconds.[1] There are many gaps in delivery of guideline-concordant therapy for cardiovascular disease.[2,3] Team-based care has become a strategy to reduce gaps in therapy.[4,5] However, small physician offices, especially in rural and small communities, lack the resources to use staff to deliver team-based care. Another strategy to assess therapy is medication therapy management typically provided by telephone, usually focused on generic agents, cost, and educating patients.[6] Medication therapy management pharmacists or nurses typically have medication lists but do not have access to medical records. A recent study of nurse-based call centers for 242 417 Medicare beneficiaries enrolled in 8 commercial disease-management programs found minimal improvements in quality-of-care measures.[7]

A systematic review of 30 trials found significant improvements in cardiovascular risk factor control with pharmacist management.[4] Investigators have called for more research to evaluate the use of pharmacists for cardiovascular disease management.[5] Our model of pharmacists embedded within medical offices, termed the physician-pharmacist collaborative model,[8,9] found that the most common intervention was intensification of medication to improve disease control.[10,11]

We developed the present model to provide virtual clinical pharmacy services from a central location. This trial was the ICARE (*I*mproved *C*ardiovascular *R*isk Reduction to *E*nhance Rural Primary Care) study with the goal to determine whether our intervention model would be implemented within private family medicine offices. The study required an outcome metric that could be used for patients with many different cardiovascular conditions or who needed preventive services. We used the Guideline Advantage (GA) criteria that were adapted from the American Heart Association's Get with the Guidelines program and developed to promote adherence to evidence-based practice guidelines in primary care settings in support of the Million Hearts initiative.[12] The primary outcome of the ICARE trial was to evaluate whether adherence to the GA metrics would be greater in practices randomized to the intervention compared with usual care.[13]

METHODS

This was a prospective, cluster-randomized trial in 12 family medicine offices in Iowa.[14] The data, analytic methods, and study materials will be made available to other researchers for purposes of reproducing the results or replicating the procedure. The Data and Resources Sharing Documents will be established at the National Heart, Lung, and Blood Institute as per their policy.[15] The study included features of a pragmatic clinical trial, except that we included 2 research visits to collect blood pressure (BP), laboratory, and other data to achieve robust data elements.[16]

Offices were identified from the Iowa Research Network, a family medicine practice-based research network directed by one of the investigators (B.T.L.).[14] Offices were screened for their ability to conduct the trial, enroll subjects, and meet quality control measures. The biostatistician (J.D.D.) randomized offices to avoid contamination that would occur if a physician had subjects in both the intervention and control groups. All subjects and physicians in a given office received either the intervention or usual care.

A study coordinator (SC) employed in each office identified, screened, and recruited subjects. In most cases, the SC was a nurse or medical assistant. Research team members traveled to each office to meet providers and train SCs on recruitment, completing case report forms, ethical conduct of research, and proper BP measurement.[17] We tracked enrollments and data quality.[18] Staffing shortages at 2 offices required that the research team provided support to perform the SC functions.

The study included 11 private physician offices and 1 Federally Qualified Health Center where care was provided by family physicians, nurse practitioners, or physician assistants. The types of practices, providers, community size, and minority population are displayed in Table 1.

Subject Recruitment

Subject recruitment followed a 2-step inclusion criteria process: first, we identified English-speaking men and women, ≥50 years of age, seen in the practice in the previous 24 months with a history of at least one of the following: (1) diabetes mellitus with hemoglobin A1c (HA1c) >7.5%, (2) hypertension, with a systolic BP (SBP) ≥150 mm Hg or a diastolic BP ≥ 90 mm Hg for subjects with uncomplicated hypertension or SBP ≥140 mm Hg SBP or diastolic BP ≥90 mm Hg for subjects with diabetes mellitus or chronic kidney disease, or (3) hypercholesterolemia with LDL (low-density lipoprotein) >110 mg/dL for subjects with a history of peripheral artery disease, coronary artery disease, stroke, transient ischemic attack, or diabetes mellitus, or >140 mg/dL in other subjects. Second, subjects

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Table 1. Medical Office Characteristics

Location	2010 Census	Minorities	Subjects Enrolled (n)
Akron	1489	0.5% Hispanic, 0.4% black	22
West Burlington	25 663	3.9% Hispanic, 4.1% black	19
Des Moines	203 433	12.0% Hispanic, 10.3% black	25
Cedar Rapids	126 326	3.3% Hispanic, 5.6% black	34
Davenport	99 685	7.3% Hispanic, 10.8% black	17
Grinnell	9218	3.2% Hispanic, 2.0% black	30
Belmond	2376	12.1% Hispanic, 0.1% black	12
Clarion	2850	15.9% Hispanic, 0.3% black	25
Knoxville	7313	1.9% Hispanic, 1.1% black	26
Newton	15 254	1.7% Hispanic, 0.7% black	35
Sioux City	82 648	16.5% Hispanic, 2.9% black	17
Riverside	993	1.2% Hispanic, 0.8% black	40

had to have a total of at least 3 disease states or risk factors that could include ≥1 of the above diseases and a history of coronary artery disease, myocardial infarction, stroke, transient ischemic attack, atrial fibrillation, peripheral vascular disease, claudication, carotid artery disease, current smoking, or meeting criteria for obesity, that is, body mass index ≥30 kg/m^2.

Exclusions included cancer with a life expectancy <24 months, pregnancy, diagnosis of primary pulmonary hypertension, inability to give informed consent, nursing home residence or diagnosis of dementia, no telephone or inability to use a telephone, inability to use the BP cuff, or plans to transfer care to a different medical office in the 12 months after recruitment.

The study and methods were approved by the University of Iowa Institutional Review Board, which served as the institutional review board of record for the study. Potential subjects were identified from patient lists, physician referral, or while in the office.[14] All subjects signed written informed consent.

BP measurements, HA1c, and LDL cholesterol were collected at baseline and 12 months regardless of whether a subject had a diagnosis of these conditions. BP was measured with the automated Omron HEM 907-XL device using a standardized technique.[8,19] Blood was drawn for lipid profile and HA1c from the usual certified laboratory used by a given office. All subjects were reimbursed $75 at baseline and follow-up.

The baseline data were collected on case report forms by SCs, sent to the research team, and entered centrally by the research team. A project manager monitored data to identify and minimize errors in data entry. Discrepancies were corrected, and SCs were educated if errors were identified.

Pharmacist Intervention

Three clinical pharmacy specialists located within the research offices of the principal investigator (B.L.C.) provided the intervention. Pharmacists were required to have a Doctor of Pharmacy (PharmD) degree and a minimum of 1 year of clinical residency (or equivalent practice experience). They had extensive experience with direct, face-to-face management of chronic medical conditions, including diabetes mellitus and cardiovascular disease in collaboration with primary care physicians. The pharmacists travelled to each intervention office to be introduced to the providers and facilitate team building before the intervention started. These sessions explored the providers' preferred methods and frequency of communication with the pharmacists based on other telemedicine studies.[20,21]

The pharmacists had access to the baseline data obtained by the SC, including BP and laboratory values. The pharmacists used this information to reconcile medications and update the medical records. Pharmacists obtained electronic medical record (EMR) access in all intervention offices that allowed for entering real-time recommendations to the providers to be placed directly into the offices' EMRs. The pharmacists wrote notes and ordered tests or medications (with physician approval) directly into the EMR in 4 offices. Pharmacists had read only access in 2 offices and made recommendations by facsimile. Pharmacists monitored blood glucose values, BP readings, laboratory values, and potential adverse events from the EMR or the patient to recommend medication changes to the medical provider. Pharmacists monitored documentation of immunization history, cancer screening, and updated alcohol and body mass index assessments, foot examinations, or measurement of microalbuminuria if appropriate (eg, diabetes mellitus). Interventions for physicians were directed at improving guideline adherence for reducing cardiovascular risk and preventive care based on the GA[12] criteria, including guidelines from the American Heart Association,[22] the diabetes mellitus guidelines,[23] the 2013 American College of Cardiology/American Heart Association cholesterol guidelines,[24] the 2014 hypertension guidelines,[25] American Heart Association/American Stroke Association guidelines on secondary prevention,[22,26–28] and screening recommendations from the American Cancer Society.[12] The metrics used in this study are displayed in the Table I in the Data Supplement.

Pharmacists were alerted from their database dashboard about gaps in guideline adherence and if physicians had not responded to a recommendation (see description in the Data Supplement).[14] The pharmacists generally did not make independent treatment changes but made recommendations to the subject's provider. However, providers in some offices gave the pharmacists authority to make medication changes or insulin adjustments and update them about the changes via the EMR. When new data or information were identified by the pharmacist from either the patient or the EMR, these data were added to the database, and the adherence to specific GA metrics was then updated as the new information became available.

Each subject was assigned to 1 of the 3 pharmacists to maintain continuity. Subjects were assigned to the pharmacists in the order they were enrolled into the study, independent of medical office. Subjects were primarily contacted by telephone and counseled by the pharmacists about beneficial health behavior changes related to tobacco use, diet, physical activity, and alcohol consumption. Medications and dosages were reconciled. The pharmacists identified problems related to poor disease control, subject misunderstandings, poor medication adherence or failure to receive preventive health services, adverse effects, drug interactions, or cost. Pharmacists provided patient education when problems were identified and developed a plan for follow-up contacts, generally every 1 to 2 weeks by telephone if there were continuing

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problems or medication issues. Intervals lengthened to every 1 to 2 months as problems resolved. The types of intervention activities are listed in the Table II in the Data Supplement.[22,26–29]

Study Aims and Outcomes

The aim of the study was to assess whether the intervention would be successfully implemented into private family physician offices. The primary outcome measure was adherence to the GA criteria as a surrogate for quality of care because each subject had varying gaps in guideline-concordant care, depending on their specific cardiovascular conditions and preventative care needs. Therefore, a proportion (0%–100% of relevant criteria met) was considered to be a more appropriate metric than the absolute number of criteria met as the possible number of criteria differed across participants. Secondary aims included control of BP, diabetes mellitus, and lipids for subjects whose conditions were not controlled at baseline.

Data Analysis

The complete discussion of the data analyses was previously published and will be summarized here.[14] There were 28 potential criteria that applied to women and 26 for men. The primary hypothesis was that adherence to the GA metric score (percent) for the intervention group would be significantly higher than the control group at 12 months. Although the outcomes were measured at the patient level, it was important to accommodate the clustering that might occur within medical offices. Therefore, our hypothesis was assessed using intent-to-treat principles and a linear mixed model in SAS Proc Mixed, adjusted for guideline adherence at baseline and including a random effect for medical office to accommodate the potential within-office correlation among subjects. Because of apparent imbalance between groups with respect to baseline LDL cholesterol, we also included that as a covariate in our mixed effects linear model, though the results were essentially identical with and without such an adjustment.

The secondary hypotheses were that mean SBP, LDL, and HA1c would be significantly lower in subjects in the intervention group compared with the control group at 12 months. The secondary hypotheses were analyzed using the same methodology as the primary outcome, except these analyses were only performed for participants whose values were uncontrolled at baseline since only those individuals required an intervention. We also analyzed the binary outcome of BP being controlled or not at 12 months post-baseline, for those whose values were out of control at baseline (ie, SBP ≥140 or diastolic BP ≥90). This analysis used a logistic mixed model (SAS Proc Glimmix) to accommodate random medical office effects.

Sample Size Justification

We previously found 40% adherence to applicable performance criteria at baseline for hypertension.[8] Others found guideline concordance of 34% for combined end points of LDL, HA1c, and SBP.[30] We used several studies that involved interventions for multiple risk factors to predict outcomes.[8,30–32] We expected baseline adherence scores would be 40%±20% and increase to 50%±20% in the control group and 60%±20% in the intervention group at 12 months (between-group effect size of 0.5 SDs). We also assumed a within-medical office intraclass correlation (ICC) of 0.005 (slightly higher than our previous study) and a dropout rate of 15%.[8] Based on these assumptions, and using formulas that accommodate random effects, we estimated that 12 medical offices with 25 subjects per office would give at least 80% power for the primary outcome, as well as all secondary outcomes, so we chose n=300 as our target sample size. Neither our power analyses nor our statistical methods adjusted for multiple comparisons across outcomes; hence, there may be a >5% chance of at least 1 type I error (ie, false-positive findings) among the secondary outcomes.

Intervention Implementation

We considered key measures of implementation to be the frequency and content of recommendations made by pharmacists and physicians' acceptance of the recommendations. We classified physician responses as accepted the recommendation in full, accepted with minor modification or rejection.[33] Medication recommendations were categorized as additions, dose change, or discontinuation. Recommendations to obtain immunizations or cancer screening were also collected and then reconciled to determine whether they were completed.

RESULTS

We recruited 302 subjects from January 2014 to October 1, 2015, and 259 (85.8%) completed the 12-month visit (Figure). Table 2 shows there were no differences in subject demographics or the GA scores at baseline between groups. For the main proxy numeric cardiovascular measures, only LDL showed a significant difference at baseline between the 2 groups, with a mean of 103.1 mg/dL in the control group and 90.6 mg/dL in the intervention group (P=0.04). Similarly, baseline hyperlipidemia was more common in the control group (94.1%) than in the intervention group (86.6%; P=0.03).[33]

Primary Aim

The median number of GA criteria that applied to subjects at baseline were 64.7% in the control group and 63.3% in the intervention group. Thus, there were ≈7 (36.7%) unmet criteria that applied per subject that pharmacists could address. There was no change in GA score from baseline to 12 months in the control group (64.7% versus 63.1%, respectively; P=0.21). There was a statistically significant improvement in the intervention group from 63.3% at baseline to 67.8% at 12 months (P=0.02). The estimated benefit of the intervention was 5.0%±2.4% (95% confidence interval=–0.5% to 10.4%; P=0.07).

Secondary Aims

We conducted a post hoc analysis to determine whether there were differences in the effect of intervention

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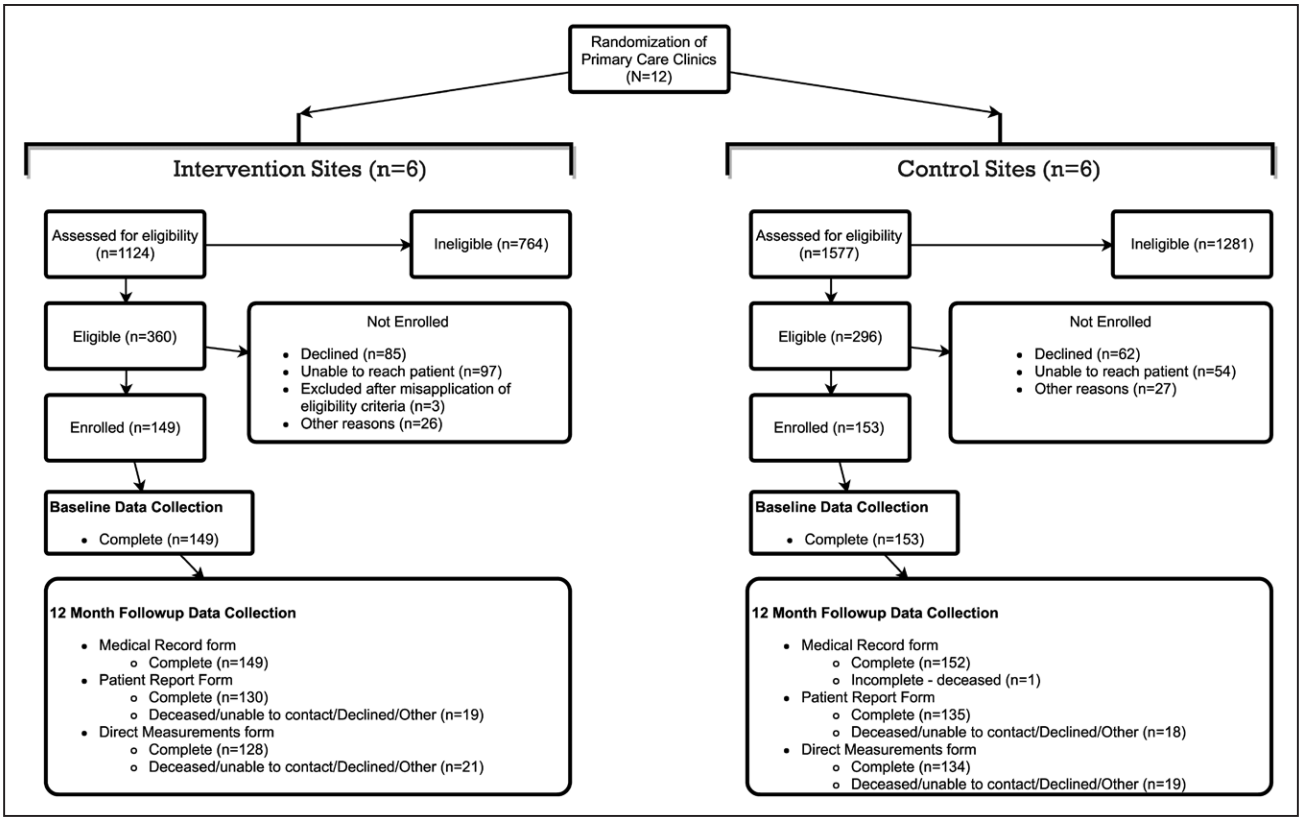

Figure. ICARE (Improved Cardiovascular Risk Reduction to Enhance Rural Primary Care) study consort flow diagram.

on guideline adherence when the pharmacists were directly responsible for making changes (n=41 subjects) compared with those subjects where a recommendation had to be made to the physician (n=108). We found that the estimated improvement in guideline adherence was more than twice as great when the pharmacist was directly responsible for changes (mean of +7.2%; SE=2.3%; $P<0.01$ compared with baseline) compared with subjects not directly managed by the pharmacist (mean of 3.2%; SE=1.3%; $P=0.02$ compared with baseline); however, the difference between these 2 intervention methods was not significant ($P=0.13$).

Table 3 displays the major individual GA metrics that applied to subjects. There were significantly more subjects in the intervention group than the control group who received appropriate statin therapy (67.5% versus 33.3%; $P<0.001$), body mass index screening (68.0% versus 37.4%; $P<0.001$), and alcohol use screening (98.0% versus 88.2%; $P<0.001$). There were several GA metrics that achieved higher rates for intervention subjects than controls (between-group comparisons at 12 months), but these were not significant including BP treatment when BP was uncontrolled (79.0% versus 70.1%; $P=0.56$), dilated eye examination (64.5% versus 58.8%; $P=0.38$), and microalbumin measured when indicated (47.7% versus 38.6%; $P=0.17$). Other metrics were similar between groups.

Table 4 displays results for subjects with uncontrolled hypertension, hyperlipidemia, and diabetes mellitus at baseline. There were consistent, modest, improvements among these 3 conditions after the intervention, but none were statistically significant. It did not appear that there was a difference in these end points depending on whether the pharmacist provided direct management or made recommendations to the physician (Table III in the Data Supplement).

Pharmacists made 331 recommendations during the intervention period, of which 294 (88.8%) were accepted in full, and 20 (6.0%) were accepted with a modified plan (Table 5). An example of a modified plan would be the physician wanting a different statin or a different dose than recommended by the pharmacist. Physicians disagreed with 15 (4.5%) recommendations, and there was no response for 2 (0.6%).

DISCUSSION

This study evaluated a comprehensive intervention to improve adherence to several cardiovascular, diabetes mellitus, cancer screening, and other preventive guidelines. The remote intervention provided by clinical pharmacists was successfully implemented and was reinforced by the support of individual office leadership to provide the pharmacists with EMR access. The pharmacists were able to rapidly communicate with physicians

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Table 2. Baseline Demographics

Characteristics (n=302)	Control (n=153), Mean (±SD) or n (%)	Intervention (n=149), Mean (±SD) or n (%)
Age (±SD)	64.1 (±8.0)	63.7 (±8.9)
Female	81 (52.9%)	69 (46.3%)
Married	81 (52.9%)	89 (59.7%)
White	144 (94.1%)	143 (96.0%)
Educational level		
≤12 y	95 (62.1%)	93 (62.4%)
Insurance		
Private	131 (85.6%)	117 (78.5%)
Medicare	10 (6.5%)	19 (12.8%)
Medicaid	10 (6.5%)	9 (6.0%)
No insurance	2 (1.3%)	4 (2.7%)
Annual income		
<$25 000	43 (28.1%)	47 (31.5%)
$25 000–$39 999	33 (21.6%)	16 (10.7%)
$40 000–$54 999	28 (18.3%)	23 (15.4%)
$55 000–$79 999	24 (15.7%)	23 (15.4%)
>$80 000	11 (7.2%)	17 (11.4%)
Refuse to answer	14 (9.2%)	23 (15.4%)
Systolic BP, mm Hg	132.9 (±21.4)	134.5 (±21.1)
Diastolic BP, mm Hg	73.8 (±13.2)	74.2 (±12.4)
Total cholesterol, mg/dL	181.4 (±44.8)	176.2 (±47.7)
HDL, mg/dL	43.7 (±12.1)	42.9 (±13.0)
LDL, mg/dL	103.1 (±39.3)	90.6 (±42.7)
Triglyceride, mg/dL	198.0 (±123.4)	211.5 (±133.0)
HA1c (%)	7.7 (±1.7)	7.7 (±1.6)
BMI, kg/m^2	36.7 (±6.9)	36.9 (±7.7)
No. of medications	5.7 (±2.7)	5.7 (±2.6)
No. of chronic conditions	5.5 (±2.0)	4.8 (±2.0)
Hypertension	143 (93.5%)	133 (89.3%)
Hyperlipidemia	144 (94.1%)	129 (86.6%)
Diabetes mellitus	125 (81.7%)	121 (81.2%)
Current smoker	34 (22.2%)	43 (28.9%)
Guideline Advantage Score	64.7% (12.8)	63.3% (13.8)

BMI indicates body mass index; BP, blood pressure; HA1c, hemoglobin A1c; HDL, high-density lipoprotein; and LDL, low-density lipoprotein.

to make recommendations through the EMR which facilitated the acceptance of the recommendations. There were modest but statistically significant improvements in guideline adherence in the intervention group. The 5% estimated improvement in the intervention group was smaller than the 10% improvement that we assumed in our initial power calculations. However, the improvement in guideline adherence was more than twice as great in subjects for whom the pharmacist could directly make medication changes and order tests compared with those in which the pharmacist simply made recommendations to physicians. These findings support new team-based models of primary care delivery but require additional exploration.

It is possible that the distant, virtual intervention was not as potent as our other studies where pharmacists were embedded within the medical office.[8,34] However, physicians' reactions were overwhelmingly positive based on the acceptance of the vast majority of recommendations. Acceptance was likely improved by initial face-to-face meetings with the physicians that allowed the pharmacists to become virtual members of the medical office team. Many physicians asked for the service to be continued after the study ended.

Telehealth interventions have achieved significant improvements in individual conditions, such as hypertension.[20,35] A Cochrane review found minimal effect of telehealth services to reduce cardiovascular risk.[36] We found consistent improvements in adherence to the GA metrics, BP, HA1c, and LDL cholesterol in intervention subjects, but these differences were modest and not significantly different from the control group. The power of this study was likely compromised because of greater interoffice variability and greater intraoffice similarity than expected (ICC=0.088 instead of the assumed 0.005). However, with an ICC of 0.088, the effective sample size was only 80, resulting in power of only 60% and an inability to see a statistically significant effect on the main outcome. A study that evaluated 21 implementation study databases found that the average ICC for outcome variables was 0.005, the same as we assumed, but outcome variables had a significantly higher ICC (0.016; $P<0.001$).[37] Our ICC was even higher which may be because of the large number of individual criteria that were process variables. We think the modest results were related to the primary outcome selected, far higher physician adherence to guideline adherence than expected at baseline, the fact that many multiple disease states were included, and that the variability between offices was greater than expected leading to low power.

One study evaluated adherence to the GA criteria in 25 practices in Utah, Illinois, Missouri, and Louisiana.[38] This study found much higher adherence to various components of the guidelines than the benchmarks established by the Million Hearts initiative, with percentages comparable to our baseline data.

Although improvements in our intervention group were modest, they may still have clinical relevance. Using the Framingham risk calculator, a simulation in our subjects would have had a reduction in cardiovascular risk of ≈25% to 50%.[39] A typical male with uncontrolled risks in our study would have reduced 10-year cardiovascular risk from ≈31% to 16%, a 48% reduction (Table IV in the Data Supplement). Risk dropped from 24% to 15% in a typical woman, a reduction of ≈25%. Given the burden of cardio-

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Table 3. Selected GA Criteria at Baseline and 12 Months

GA Criteria (Recommendations and Issues Addressed)	12 Months				
	Control (n)	Intervention (n)	Control (n)	Intervention (n)	*P* Value
Blood pressure treatment (≥2 meds w/ uncontrolled hypertension)	87.5% (21/24)	76.0% (19/25)	70.1% (12/17)	79.0% (15/19)	0.5631
Uncontrolled hypertension	50% (75/150)	51.0% (75/147)	30.2% (45/149)	30.4% (45/148)	0.9695
Uncontrolled DM	86.1% (99/115)	86.5% (96/111)	79.0% (90/114)	69.2% (74/107)	0.0965
Symptom and activity assessment	96.4% (54/56)	79.6% (43/54)	93.7% (59/63)	87.1% (54/62)	0.2136
Advising smokers to quit	79.7% (122/153)	73.2% (109/149)	92.1% (140/152)	87.9% (131/149)	0.2255
Tobacco replacement therapy recommended	0 (0/31)	6.4% (3/47)	3.1% (1/32)	11.6% (5/43)	0.1794
Alcohol use screening	76.5% (117/153)	91.3% (136/149)	88.2% (134/152)	98.0% (146/149)	0.0008
Appropriate statin recommended	32.9% (45/137)	33.1% (46/139)	33.3% (42/126)	67.5% (83/123)	<0.001
Dilated eye examination recommended	76.5% (88/115)	66.7% (74/111)	58.8% (67/114)	64.5% (69/107)	0.3829
Diabetic foot examination	90.4% (104/115)	76.6% (85/111)	78.1% (89/114)	76.6% (82/107)	0.7989
Microalbumin order recommended	57.4% (66/115)	73.0% (81/111)	38.6% (44/114)	47.7% (51/107)	0.1736
BMI: screening and follow-up	50.7% (76/150)	66.9% (95/142)	37.4% (49/131)	68.0% (83/122)	<0.001

BMI indicates body mass index; DM, diabetes mellitus; and GA, Guideline Advantage.

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vascular disease, these reductions may be important. These reductions in cardiovascular risk are similar to a pharmacy-based intervention in Canada[40] and suggest that implementation of this intervention should focus on subjects at higher risk.

There were several limitations to this study, and we learned important lessons that may improve intervention effectiveness and plan future research. First, many patients had limited gaps in therapy for the pharmacists to address. This intervention should be more potent if physicians refer patients to this service who have uncontrolled conditions and are in greater need of intervention. Second, the intervention would be enhanced if the physician and pharmacist have formal collaborative practice agreements that allow pharmacists to initiate or modify therapy and order various laboratory tests. Collaborative practice agreements are recognized by state boards of medicine and pharmacy in most of the states in the United States. Although our pharmacists were granted this authority for some subjects, our intervention would have been more potent with formal agreements. Third, the GA metrics are important benchmarks for medical offices but may be suboptimal as a research outcome because of the large number of criteria, many of which are process variables.[12] Providers had good performance, and baseline adherence was higher than expected which gave less room for improvement. However, the estimated effect size was still ≈0.5 SDs, which is of medium magnitude, consistent with our target treatment effect. We expected multiple conditions to be uncontrolled in many patients which often was not the case because of better-than-expected control rates. The relatively small numbers of subjects with any given uncontrolled condition limited power for these individual end points and also improved the overall GA scores. Future research with this model should focus on a few selected, uncontrolled condi-

Table 4. Analyses of Secondary Outcomes for Subjects With Uncontrolled Baseline Measures*

Outcome	Baseline Eligibility Rule	12-Month Data Source	n	Estimated Effect (CI), Intervention Minus Control	*P* Value
SBP, mm Hg	Uncontrolled BP (standard)	Research visit only	99	−1.78 (−8.73, 5.18)	0.61
		Research visit or chart	115	−2.47 (−8.99, 4.05)	0.46
	Uncontrolled BP (age adjusted)	Research visit only	78	−4.01 (−11.9, 3.87)	0.31
		Research visit or chart	90	−4.26 (−11.69, 3.17)	0.26
LDL, mg/dL	LDL ≥100 mg/dL	Research visit only	108	−2.8 (−22.9, 17.3)	0.73
	LDL ≥130 mg/dL	Research visit only	51	−7.1 (−30.7, 16.4)	0.55
HA1c	HA1c ≥7.5%	Research visit only	137	−0.54 (−1.10, 0.02)	0.06

BP indicates blood pressure; CI, confidence interval; HA1c, hemoglobin A1c; LDL, low-density lipoprotein; and SBP, systolic blood pressure.

*Standard definition of uncontrolled was SBP >140 or diastolic BP >90 for all ages; age-adjusted required SBP to be >150 or diastolic BP >90 for those ≥60 years of age. The most recent chart values or research values were used for missing data at 12 months. Only subjects who met those individual criteria or the LDL or HA1c values in the table are included in these analyses.

Table 5. Intervention Group Pharmacy Recommendations and Acceptance by Providers

Recommendation	Provider Outcome	Count	Subtotal and % Accepted Completely	% of All Recommendations
Increase dose	Accepted	95	104 (91.3%)	31.4
	Accepted with minor modification	6		
	Disagreed	3		
Initiate new medication	Accepted	80	92 (87.0%)	27.8
	Accepted with minor modification	7		
	Disagreed	5		
Change medication	Accepted	52	59 (88.1%)	17.8
	Accepted with minor modification	2		
	Disagreed	4		
	No response	1		
Discontinue medication	Accepted	16	22 (72.7%)	6.6
	Accepted with minor modification	5		
	No response	1		
Order laboratory	Accepted	20	22 (90.9%)	6.6
	Disagreed	2		
Decrease dose	Accepted	14	14 (100%)	4.2
Pneumonia vaccination	Accepted	6	6 (100%)	1.8
Foot examination	Accepted	5	5 (100%)	1.5
Colorectal cancer screening	Accepted	5	5 (100%)	1.5
Influenza vaccination	Accepted	1	2 (50%)	0.6
	Disagreed	1		
	Total	331	331 (88.8%)	100

tions that should provide more power and the ability to better assess the cost-effectiveness of this intervention. Fourth, there is a possibility that selection bias occurred based on the strategy SCs used to identify subjects. However, we conducted a comprehensive analysis and found that the probability of selection bias was low.[41] Finally, our study included few minority subjects, and an evaluation of greater numbers of subjects from minority groups would increase the generalizability of the trial.

CONCLUSIONS

We found that providers in intervention practices accepted the vast majority of recommendations for cardiovascular and preventive care provided by centralized pharmacists. The intervention was enhanced by gaining remote access to EMRs. Although the intervention led to modest improvements in cardiovascular risk factors, these were not statistically significant because of more interoffice variability than anticipated. In addition, we expected baseline guideline adherence to be 40%, but the observed adherence was >60%, which reduced the ability to markedly improve guideline adherence. Future implementation of this care model should focus on patients with uncontrolled conditions at highest risk for cardiovascular events.

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Cluster-Randomized Trial to Evaluate a Centralized Clinical Pharmacy Service in Private Family Medicine Offices

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Supplemental Material

A cluster randomized trial to evaluate a centralized clinical pharmacy service in private family medicine offices

Barry L. Carter

Short title: Pharmacist model to improve guideline adherence

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Table S1: Guideline Advantage Criteria Comparison (Based on 2011 and 2013 Fact Sheets)

Criteria	GA 2011	GA 2013	GA used
Atrial Fib	Assessment of thrombotic risk factors	NOT addressed	2011
	Chronic anticoagulation therapy	NOT addressed	2011
	Monthly INR for patients on warfarin	NOT addressed	2011
Mammography	Women age 40-69 screened in last 24 months	Same	2013
Colorectal	Age 50-75 with appropriate cancer screening	Same	2013
Cervical	Women age 21-69 who had Pap screen in last 2 years	Women age 21-63 who had Pap screen in last 3 years	2013
CAD	% 18 or older prescribed antiplatelet therapy	% 18 or older prescribed aspirin or clopidogrel	2013
	CAD and prior MI prescribed B-Blocker	NOT addressed	2011
	CAD and DM or CHF prescribed an ACE/ARB	NOT addressed	2011
	Prescribed lipid lowering agent for LDL	Same	2013
	NOT addressed	LDL test in previous 12 months	2013
	NOT addressed	LDL < 100mg/dl	2013
	Evaluated for level of activity and angina symptoms	NOT addressed	2011
	Angina free or on at least 2 anti-anginal medications	NOT addressed	2011
	Referred to cardiac rehab if had MI in last 2 years	NOT addressed	2011
	NOT addressed	BP screen	2013
	NOT addressed	BP < 140/90mmHg	2013

	Same	BP > 140/90mmHg and prescribed 2 anti-hypertensives	2013
	NOT addressed	CAD and tobacco screen in last 12 months	2013
	Asked about smoking status 1 or more times in last 24 months	Smokers who have received smoking cessation intervention in last 24 months	2013
	Addressed in "Preventative Care"	BMI calculated in last 6 months with a follow up plan documented if outside parameters.	2013
	NOT addressed	BMI outside parameters	2013
DM	Age 18-75 with A1c >9.0 OR 18-64 with A1c <7.0	Age 18-75 with A1c <7.0, 7.0-7.9, 8.0-9.0 and >9.0	2013
	NOT addressed	A1c in last 12 months	2013
	Age 18-75 with LDL < 100mg/dl	Same	2013
	NOT addressed	Age 18-75 and LDL test in last 12 months	2013
	Age 18-75 with BP <140/80mmHg	Same	2013
	NOT addressed	Age 18-75 and BP screen in last 12 months	2013
	Urine albumin screening in last 12 months	Same	2013
	Age 18-75 who have had a dilated eye exam	NOT addressed	2011
	Age 18-75 who have had a foot exam	NOT addressed	2011
CHF	Qualitative or quantitative results of LVF assessment	NOT addressed	2011
	ACE or ARB for patients with LVEF < 40%	NOT addressed	2011
	Beta-Blocker for patients with LVEF < 40%	NOT addressed	2011
HTN	NOT addressed	Age 18-85 with a BP screen in last 12 months	2013
	NOT addressed	Age 18-85 with BP < 140/90mmHg	2013
PAD	Smokers who have had a cessation intervention	Only addressed as preventative care, not PAD specific	2011

Preventative Care	Unhealthy alcohol screening	NOT addressed	2011
	Age 50 and older with Influenza shot	Age 18 and older with Influenza shot	2013
	Age 65 and older with Pneumonia shot	Age 18 years and older who are tobacco users who have received Pneumonia vaccine	2013
	Controlled LDL OR prescribed at least 1 lipid lower med at max dose tolerated	Age 18 and older with LDL test in last 5 years	Both
	Counseling to obtain and maintain IBW in last 2 years	Age 18 and older with a calculated BMI in last 6 months	2013
	NOT addressed	Age 18 and older with a calculated BMI in last 6 months outside of parameters	2013
	NOT addressed	Age 18 and older with a calculated BMI in last 6 months outside of parameters and a documented f/up plan	2013
	BP measurement in ALL patients	Age 18 and older with BP measurement in last 2 years	2013
	Advised to use ASA	NOT addressed	2011
	Smokers age 18 and older advised to quit	Same	2013
	NOT addressed	Age 18 and older screened for tobacco use	2013
	NOT addressed (only under PAD)	Age 18 and older who are tobacco users and have received cessation intervention in the last 24 months	2013
Asthma	NOT addressed	Mild, moderate or severe asthma prescribed long-term control med or appropriate alternative	2013

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Shaded criteria are the elements used in the present study. There were 26 potential criteria that could apply to males and 28 criteria that could apply to females.

The Pharmacist Intervention

The pharmacist maintained records in their personal health record (PHR) account, could track subjects' self-monitored data (e.g. BP, blood glucose, lipids), and could reconcile and update medications and recommend preventive services (e.g. immunizations). The PHR was a unique database management system developed prior the present trial by several investigators (EAC, BG) and the PHR is being used after the completion of the present trial.

The PHR allowed pharmacists to maintain records of medications, laboratory data, self-monitored data (e.g. blood glucose or BP), pharmacist progress notes, recommendations made to physicians and subjects and physician acceptance. The PHR provided the pharmacists with alerts to schedule follow-up with subjects and problems to address and a complete accounting of time required to perform the intervention for each subject. The types of interventions are displayed in Table S2.

Table S2. Selected Examples of pharmacists' recommendations*

Patient Circumstances	Typical CVRS Pharmacist Recommendation
Condition not controlled and patient not adhering to regimen	Address reasons for non-adherence, stages of change, adjust regimen, monitor.
Patient continues to smoke	Address readiness to quit, refer to tobacco quit line, start medications.
Patients with Type II diabetes not taking aspirin, ACEI (or ARB), statin, metformin	Add aspirin 75-162mg , add ACEI (or ARB if adverse effects to ACEI), add statin, add/or metformin if no contraindications and adjust to proper dose
No documented foot or eye exam, no microalbumin.	Recommend foot or eye exam and measure urinary microalbumin
CAD, MI and not taking aspirin, beta blocker or ACE inhibitor	Begin aspirin 75-162 mg and/or beta blocker if no contraindications. ACEI (or ARB) if no contraindications
Elevated cholesterol risk	Start statin or increase dose. Add additional agents to treat specific lipid disorder and to achieve goal.
DM with HA1c $\geq$ 7%	Optimize therapy as necessary.
SBP or DBP above goal and adhering to lifestyle and/or medications	Add thiazide-like diuretic if not part of regimen, otherwise add synergistic second-line drug, reinforce lifestyle modification
Important drug interactions or adverse drug events	Adjust medication regimen to eliminate or minimize drug problems
Overdue cancer screening (e.g. colonoscopy, mammography)	Recommend appropriate screening.

*** -** Note, examples are not all inclusive. ACEI – angiotensin converting enzyme inhibitor; ARB – angiotensin receptor blocker, CAD- coronary artery disease; MI – myocardial infarction, DM – diabetes mellitus, HA1c – hemoglobin A1c, SBP – systolic blood pressure, DBP-diastolic blood pressure.

Table S3 displays the effect of the pharmacist intervention when the pharmacists directly managed patients and made medication changes compared to when they had to make recommendations to the physician. The improvement in the guideline score was over twice as great when the pharmacists made changes (+7.2) compared to when the pharmacist did not directly manage the patient (+3.2). Both were significantly better than baseline but not different from each other perhaps related to the small sample size.

The improvement in BP and HA1c in these two groups were difficult to interpret due to small numbers of subjects with uncontrolled values at baseline.

Table S3: Comparison of Subjects directly managed by pharmacists (12 month minus baseline)*

	Pharmacist directly managed			**Pharmacist did not directly manage**		
	N pairs	**Mean (SE)**	**p-values**	**N pairs**	**Mean (SE)**	**p-values**
Change in guideline adherence score	41	+7.2 (2.3)	<0.01	108	+3.2 (1.3)	0.02
BP not controlled at baseline						
Systolic BP (mmHg)	22	-12.4 (3.7)	<0.01	36	-16.2 (4.1)	<0.01
Diastolic BP (mmHg)	22	-6.6 (2.5)	0.02	36	-7.8 (1.8)	<0.01
Diabetes when HA1c>7.5 at baseline						
HbA1c	17	-0.81 (0.20)	<0.01	62	-0.74 (0.22)	<0.01

* - Change from baseline for subjects in the intervention group. Subjects were included for the BP and diabetes analysis if these were uncontrolled at baseline. “Directly managed” indicates medication changes made by the pharmacist. Pharmacists made recommendation to physicians to make changes for subjects “not directly managed.” P values are compared to baseline. SE = standard error.

Case Simulations:

Examples of the typical improvement in risk factors were used to estimate changes in the Framingham risk score following the intervention while holding age constant (Table S4). The effect of the intervention would be associated with a 48% reduction in 10 year cardiac risk for male and a 25% reduction in 10 year cardiac risk for the female examples.

Table S4: Case Simulations Keeping Age constant

Male Aged 64 years at baseline		
Risk Factor	Value	Points
Age	64 years	5
Total Cholesterol	210 mg/dL	1
HDL	43 mg/dL	1
Treated SBP	143 mm Hg	2
Smoker	No	0
Diabetes	Yes	2
Point Total		11
Estimated 10 year risk		31%
Male Aged 64 years following intervention		
Risk Factor	Value	Points
Age	64 years	5
Total Cholesterol	198 mg/dL	0
HDL	45 mg/dL	0
Treated SBP	138 mm Hg	1
Smoker	No	0
Diabetes	Yes	2
Point Total		8
Estimated 10 year risk		16%

Female Aged 64 years at baseline		
Risk Factor	Value	Points
Age	64 years	8
Total Cholesterol	210 mg/dL	1
HDL	45 mg/dL	1
Treated SBP	143 mm Hg	2
Smoker	No	0
Diabetes	Yes	4
Point Total		16
Estimated 10 year risk		24%
Female Aged 64 years following intervention		
Risk Factor	Value	Points
Age	64 years	8
Total Cholesterol	198 mg/dL	0
HDL	50 mg/dL	0
Treated SBP	138 mm Hg	1
Smoker	No	0
Diabetes	Yes	4
Point Total		13
Estimated 10 year risk		15%

www.medcalc.com[1]

1. Coronary Heart Disease Risk Calculator. http://www.medcalc.com/heartrisk.html Accessed January 8 2018.